

# Systems Evaluations — Slide 1

## Systems Evaluations

### Quality Control for AI Agents

By the end of this session, you will: ✓ Understand why evaluation differs from testing ✓ Design and implement evaluation rubrics ✓ Run hybrid (automated + manual) evaluations ✓ Document decisions for reproducibility

# Systems Evaluations — Slide 2

## The Problem

You built an AI agent. It passes all tests. But in production...

## What Happened

✗ Agent hallucinates facts ■ ✗ Responses unhelpful ■ ✗ Users complain ■ ✗ No way to measure quality

## Why Testing Failed

✓ Code works ■ ✓ Edge cases pass ■ ✓ Performance is fast ■ ✗ But user experience?

Tests verify code. Evaluations verify real-world value.

Testing: Does the code run? ■ Evaluation: Does it solve the user's problem?

# Systems Evaluations — Slide 3

What is Evaluation?

Systematic measurement of whether an agent achieves its goals

→

Does it work?

Produces output without errors

→

Meets standards?

Output meets quality bar

→

Where fails?

Which use cases break?

→

How improve?

What specific changes help?

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Three Approaches

## **AUTOMATED**

Fast: Seconds ■ Scalable: 1000s ■ Repeatable ■ Cheap ■ Limited nuance

## **MANUAL**

Nuanced ■ Context-aware ■ Catches edge cases ■ Slow: Hours ■ Expensive

## **HYBRID ■**

Automated: First pass ■ Manual: Spot-check ■ Cost-effective ■ Catches both issues ■ Best approach

Industry standard: Run automated, then manual review on sample

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The 5-Step Framework

## 1. DEFINE

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What goals matter?

Quality, helpfulness, accuracy, speed

## 2. DESIGN

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How measure?

Rubrics, metrics, scoring system

## 3. BUILD

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Create rubric

Criteria → Levels → Descriptors

## 4. TEST

### 4. TEST

Pilot run

Test on 50-100 samples first

## 5. SCALE

### 5. SCALE

Full eval

Apply to complete dataset

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Real Rubric Example: Fact Accuracy

Scenario: Evaluating if agent gives correct information

**EXEMPLARY**

**EXEMPLARY**

Level 4

All facts correct, proper citations, complete answer

**PROFICIENT**

**PROFICIENT**

Level 3

No major errors, 90%+ correct, mostly complete

**DEVELOPING**

**DEVELOPING**

Level 2

Some errors (1-2), partial answer, confusing

**NEEDS WORK**

**NEEDS WORK**

Level 1

Major errors or hallucinations, incomplete

Notice: Specific descriptors → No ambiguity → Consistent scoring

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## Pitfalls & How to Avoid Them

- Vague criteria

- ✓ Use measurable terms

'good' → 'factually accurate'

- Rater bias

- ✓ Structured rubric

Objective scoring, not gut feeling

- Too narrow sample

- ✓ Representative data

Test on diverse cases first

- No documentation

- ✓ Log every decision

Why did you score it 3/4?

- Single rater

- ✓ Multiple judges

Compare results, calibrate

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Exercise: Design Your Rubric

Scenario: Evaluating customer support responses

What makes a GOOD support response? (Next 5 minutes: Discuss with partner)

What's the main goal? (Speed? Accuracy? Empathy?) How would you measure each criterion? What's exemplary vs. needs work? What would bias your judgment? How would you document your decision?



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When Done Right: Real Impact

Before Evaluation

✗ Agent seems fine ■ ✗ Users complain ■ ✗ 40% satisfaction ■ ✗ No visibility

After Evaluation

✓ Identified 3 bugs ■ ✓ Improved quality ■ ✓ 87% satisfaction ■ ✓ Measurable progress

Structured evaluation → Visible improvements → Data-driven decisions

Example: Query about product pricing Before: "Response is unhelpful" (Vague)

After: "Misses 2 of 5 required price points" (Measurable) → Fix it!

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Checklist: Ready to Evaluate?

- Define what 'good' means for your agent (1-2 goals)
- Design 1 rubric with 4 levels + descriptors
- Pick 20-30 representative test cases
- Have 2 people score the same 5 samples
- Compare results - do they match? (Calibrate if not)
- Document scoring decisions + rationale
- Run on full test set
- Analyze results → What patterns?
- Plan improvements based on findings

Start simple. You can always refine.

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## What You've Learned

1. Evaluation  $\neq$  Testing — Measure real-world value, not just code correctness
2. Structured rubrics eliminate bias and ensure consistency
3. Hybrid approach works: Automated screening + Manual verification
4. Start small: Pilot on 20-30 samples before scaling to thousands
5. Document everything: Every decision is part of your quality audit trail

# Systems Evaluations — Slide 12

Questions?

Next: Pick one of your agents and design an evaluation Send results to [aroma.tahir@taleemabad.com](mailto:aroma.tahir@taleemabad.com) We'll review and refine together